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Section 2.5

Matrix Factorizations

Matrix Factorizations

A factorization of a matrix A is an equation that expresses A as the product of two or more matrices.

We will consider one common factorization, the LU factorization.

The LU Factorization

If A is an $m \times n$ matrix that can be row reduced to echelon form **without row interchanges**, A can be written in the form

$$A = LU$$

where L is an $m \times m$ lower triangular matrix with 1s on the main diagonal and U is an $m \times n$ echelon form of A .

The LU Factorization

The LU factorization can be used to solve $A\mathbf{x} = \mathbf{b}$.

First, rewrite the equation as

$$L(U\mathbf{x}) = \mathbf{b}$$

Then, since $U\mathbf{x}$ is a vector, we can let

$$U\mathbf{x} = \mathbf{y}$$

and solve the pair of equations

$$L\mathbf{y} = \mathbf{b}$$

$$U\mathbf{x} = \mathbf{y}$$

Example 1

Solve the equation $A\mathbf{x} = \mathbf{b}$ using the LU factorization given below.

$$A = \begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 3 & -5 \\ 0 & -2 & 2 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\mathbf{b} = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}$$

Finding an LU Factorization

1. Reduce A to an echelon form U by a sequence of row replacement operations, if possible.
(We will not consider the case where this is not possible.)
2. Place entries in L such that the same sequence of row replacement operations reduces L to I

Example 2

Find an LU factorization of

$$A = \begin{bmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{bmatrix}$$

Example 3

Find an LU factorization of

$$A = \begin{bmatrix} 2 & -4 & -2 & 3 \\ 6 & -9 & -5 & 8 \\ 2 & -7 & -3 & 9 \\ 4 & -2 & -2 & -1 \\ -6 & 3 & 3 & 4 \end{bmatrix}$$

Numerical Notes

For an $n \times n$ dense matrix A with $n \geq 30$:

- Computing an LU factorization of A takes about $2n^3/3$ flops, about the same as row reducing $[A \quad \mathbf{b}]$.
- Finding A^{-1} requires about $2n^3$ flops.

Numerical Notes

For an $n \times n$ dense matrix A with $n \geq 30$:

- Solving $L\mathbf{y} = \mathbf{b}$ and $U\mathbf{x} = \mathbf{y}$ requires about $2n^2$ flops.
- Multiplication of $\mathbf{b}$ by A^{-1} also requires about $2n^2$ flops, but may result in a less accurate answer than that obtained by LU due to roundoff errors.

Numerical Notes

If A is sparse, then L and U may also be sparse whereas A^{-1} is likely to be dense. In this case, we are much better off using LU than A^{-1}
